



CYBER SECURITY INTERNSHIP PROGRAM

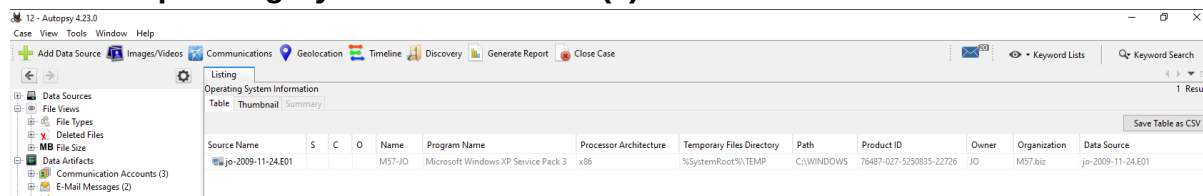
Muhammad Zubair Yaqoob

ROLL NUMBER: CSI-B1-050

Part A — Triage and Subject Profiling

Case Setup and System Baseline

In the left navigation panel (Tree Viewer), expand the **Data Artifacts** tree. Click on **Operating System Information (1)**.



The screenshot shows the Autopsy 4.23.0 interface. The left sidebar shows the 'Data Artifacts' tree expanded, with 'Operating System Information (1)' selected. The main pane displays a table with the following data:

Source Name	S	C	O	Name	Program Name	Processor Architecture	Temporary Files Directory	Path	Product ID	Owner	Organization	Data Source
jo-2009-11-24.E01				M57-JO	Microsoft Windows XP Service Pack 3	x86	%SystemRoot%\TEMP	C:\WINDOWS	76487-027-5250835-22726	JO	M57.biz	jo-2009-11-24.E01

Hostname (System Name): M57-JO

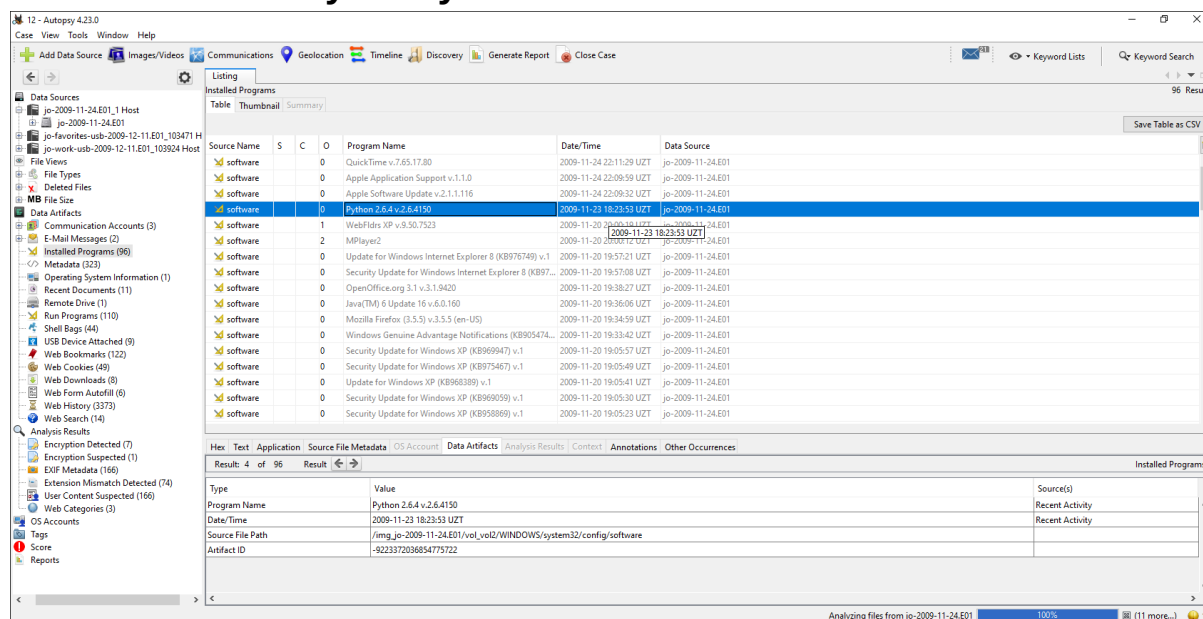
Operating System: Microsoft Windows XP Service Pack 3

Architecture: x86

Timezone: GMT

Last Known Login: *Note: Autopsy's automated OS Accounts extraction module was unable to parse this data, likely due to the legacy architecture of Windows XP. Manual registry analysis is required.*

Software Inventory Analysis



The screenshot shows the Autopsy 4.23.0 interface with the 'Software Inventory' table. The table lists various installed programs. The following table represents the data shown in the screenshot:

Source Name	S	C	O	Program Name	Date/Time	Data Source
software	0	0	0	QuickTime v.7.65.17.80	2009-11-24 22:11:39 UZT	jo-2009-11-24.E01
software	0	0	0	Apple Application Support v.1.1.0	2009-11-24 22:09:59 UZT	jo-2009-11-24.E01
software	0	0	0	Apple Software Update v.2.1.1.116	2009-11-24 22:09:32 UZT	jo-2009-11-24.E01
software	1	0	0	Python 2.6.4 v2.6.4.150	2009-11-23 18:23:53 UZT	jo-2009-11-24.E01
software	1	0	0	WebPiles XP v.8.50.7523	2009-11-20 19:02:26 UZT	jo-2009-11-24.E01
software	2	0	0	MPlayer2	2009-11-23 18:23:53 UZT	jo-2009-11-24.E01
software	0	0	0	Update for Windows Internet Explorer 8 (KB977480) v.1	2009-11-20 19:57:21 UZT	jo-2009-11-24.E01
software	0	0	0	Security Update for Windows Internet Explorer 8 (KB97...	2009-11-20 19:57:08 UZT	jo-2009-11-24.E01
software	0	0	0	OpenOffice.org 3.1 v.3.1.9420	2009-11-20 19:38:27 UZT	jo-2009-11-24.E01
software	0	0	0	Java(TM) 6 Update 16 v.6.0.160	2009-11-20 19:36:06 UZT	jo-2009-11-24.E01
software	0	0	0	Mozilla Firefox (3.5.5) v.3.5.5 (en-US)	2009-11-20 19:34:59 UZT	jo-2009-11-24.E01
software	0	0	0	Windows Genuine Advantage Notifications (KB905474...	2009-11-20 19:33:42 UZT	jo-2009-11-24.E01
software	0	0	0	Security Update for Windows XP (KB969947) v.1	2009-11-20 19:05:57 UZT	jo-2009-11-24.E01
software	0	0	0	Security Update for Windows XP (KB975467) v.1	2009-11-20 19:05:49 UZT	jo-2009-11-24.E01
software	0	0	0	Update for Windows XP (KB968389) v.1	2009-11-20 19:05:41 UZT	jo-2009-11-24.E01
software	0	0	0	Security Update for Windows XP (KB969059) v.1	2009-11-20 19:05:30 UZT	jo-2009-11-24.E01
software	0	0	0	Security Update for Windows XP (KB958869) v.1	2009-11-20 19:05:23 UZT	jo-2009-11-24.E01

Critical Anomaly:

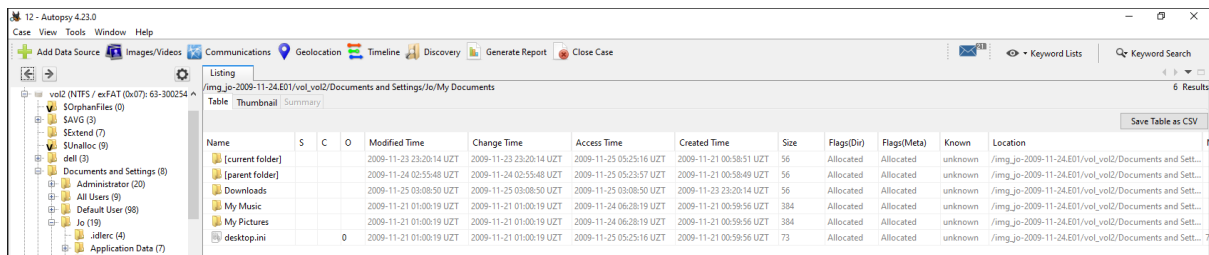
Python 2.6.4 was installed on **November 23, 2009, at 18:23:53 UZT.**

Secondary Anomaly:

Apple Application Support and **QuickTime** were installed on **November 24, 2009, around 22:09 UZT.**

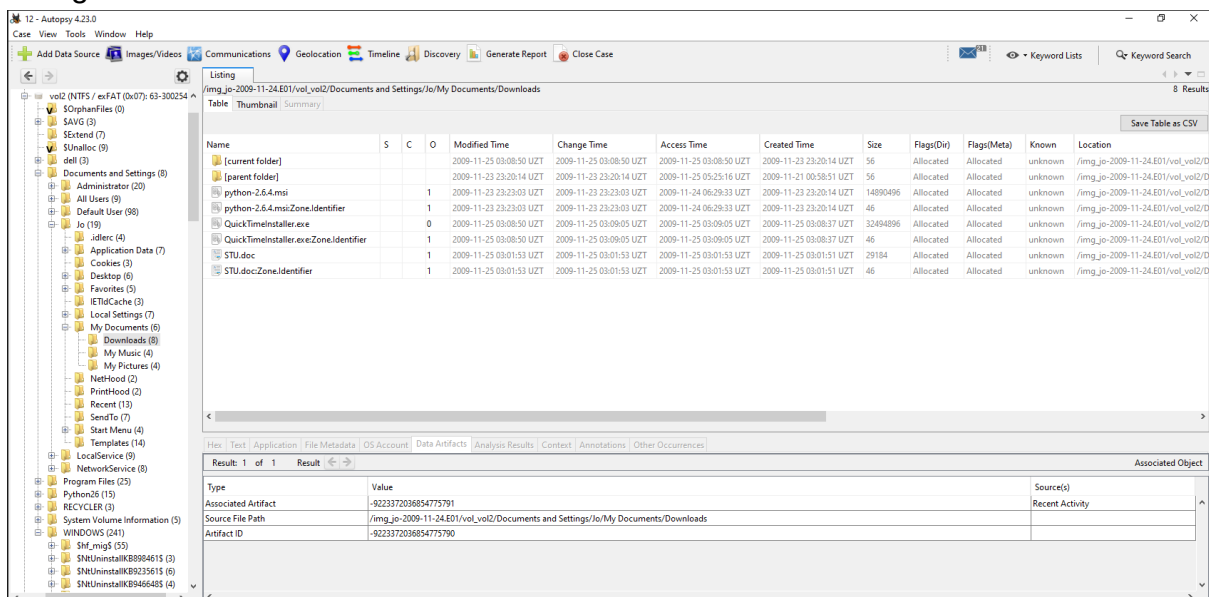
Subject Profiling

To Examine jo's profile we need to navigate to the **Data Sources > jo-2009-11-24.E01 > vol2 > Documents and Settings > jo > My Documents**



Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location
[current folder]				2009-11-23 23:20:14 UZT	2009-11-23 23:20:14 UZT	2009-11-25 05:25:16 UZT	2009-11-21 00:58:51 UZT	56	Allocated	Allocated	unknown	/img_jo-2009-11-24.E01/vol2/Documents and Sett...
[parent folder]				2009-11-24 02:55:48 UZT	2009-11-24 02:55:48 UZT	2009-11-25 05:23:57 UZT	2009-11-21 00:58:49 UZT	56	Allocated	Allocated	unknown	/img_jo-2009-11-24.E01/vol2/Documents and Sett...
Downloads				2009-11-25 03:08:50 UZT	2009-11-25 03:08:50 UZT	2009-11-25 03:08:50 UZT	2009-11-23 23:20:14 UZT	56	Allocated	Allocated	unknown	/img_jo-2009-11-24.E01/vol2/Documents and Sett...
My Music				2009-11-21 01:00:19 UZT	2009-11-21 01:00:19 UZT	2009-11-24 06:28:19 UZT	2009-11-21 00:59:56 UZT	384	Allocated	Allocated	unknown	/img_jo-2009-11-24.E01/vol2/Documents and Sett...
My Pictures				2009-11-21 01:00:19 UZT	2009-11-21 01:00:19 UZT	2009-11-24 06:28:19 UZT	2009-11-21 00:59:56 UZT	384	Allocated	Allocated	unknown	/img_jo-2009-11-24.E01/vol2/Documents and Sett...
desktop.ini			0	2009-11-21 01:00:19 UZT	2009-11-21 01:00:19 UZT	2009-11-25 05:25:16 UZT	2009-11-21 00:59:56 UZT	73	Allocated	Allocated	unknown	/img_jo-2009-11-24.E01/vol2/Documents and Sett...

Navigate to the Downloads folder



Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location
[current folder]				2009-11-25 03:08:50 UZT	2009-11-25 03:08:50 UZT	2009-11-25 03:08:50 UZT	2009-11-23 23:20:14 UZT	56	Allocated	Allocated	unknown	/img_jo-2009-11-24.E01/vol2/Doc...
[parent folder]				2009-11-23 23:20:14 UZT	2009-11-23 23:20:14 UZT	2009-11-25 05:25:16 UZT	2009-11-21 00:58:51 UZT	56	Allocated	Allocated	unknown	/img_jo-2009-11-24.E01/vol2/Doc...
python-2.6.4.msi			1	2009-11-23 23:23:03 UZT	2009-11-23 23:23:03 UZT	2009-11-24 06:29:33 UZT	2009-11-23 23:20:14 UZT	14890496	Allocated	Allocated	unknown	/img_jo-2009-11-24.E01/vol2/Doc...
python-2.6.4.msi:Zone.Identifier			1	2009-11-23 23:23:03 UZT	2009-11-23 23:23:03 UZT	2009-11-24 06:29:33 UZT	2009-11-23 23:20:14 UZT	46	Allocated	Allocated	unknown	/img_jo-2009-11-24.E01/vol2/Doc...
QuickTimeInstaller.exe			0	2009-11-25 03:08:50 UZT	2009-11-25 03:09:05 UZT	2009-11-25 03:09:05 UZT	2009-11-25 03:08:57 UZT	32494896	Allocated	Allocated	unknown	/img_jo-2009-11-24.E01/vol2/Doc...
QuickTimeInstaller.exe:Zone.Identifier			1	2009-11-25 03:08:50 UZT	2009-11-25 03:09:05 UZT	2009-11-25 03:09:05 UZT	2009-11-25 03:08:57 UZT	46	Allocated	Allocated	unknown	/img_jo-2009-11-24.E01/vol2/Doc...
STU.doc			1	2009-11-25 03:01:53 UZT	2009-11-25 03:01:53 UZT	2009-11-25 03:01:53 UZT	2009-11-25 03:01:51 UZT	29184	Allocated	Allocated	unknown	/img_jo-2009-11-24.E01/vol2/Doc...
STU.doc:Zone.Identifier			1	2009-11-25 03:01:53 UZT	2009-11-25 03:01:53 UZT	2009-11-25 03:01:53 UZT	2009-11-25 03:01:51 UZT	46	Allocated	Allocated	unknown	/img_jo-2009-11-24.E01/vol2/Doc...

Type	Value	Source(s)
Associated Artifact	-9223372036854775791	Recent Activity
Source File Path	/img_jo-2009-11-24.E01/vol2/Documents and Settings/jo/My Documents/Downloads	
Artifact ID	-9223372036854775790	

The user exhibits anomalous access patterns for a non-technical employee. Artifact analysis of the **Downloads** directory reveals the direct internet download of standalone executables, specifically **python-2.6.4.msi** (verified by the **:Zone.Identifier** Alternate Data Stream).

Secondary Artifact: An unexplained document named **STU.doc** was also located in the Downloads directory alongside the installer.

Hunting the Download URL

Navigate to the **Data Artifacts** and then **Web History**

Source Name	S	C	O	URL	Date Accessed	Title	Program Name	Domain	Data Source	Referrer URL
places.sqlite		2		http://en-us.start3.mozilla.com/firefox/clients/firefox-...	2009-11-23 23:17:46 UZT	firefox	Firefox Analyzer	mozilla.com	jo-2009-11-24.E01	
places.sqlite		3		http://www.google.com/firefox/client-firefox-a&site=...	2009-11-23 23:17:43 UZT	Mozilla Firefox Start Page	Firefox Analyzer	google.com	jo-2009-11-24.E01	http://en-us.start3.mozilla.com/firefox/clients/...
places.sqlite		3		http://www.google.com/search?q=mozilla&btnG=...	2009-11-23 23:17:55 UZT	mozilla - Google Search	Firefox Analyzer	google.com	jo-2009-11-24.E01	http://www.google.com/firefox/clients/...
places.sqlite		1		http://wiki.github.com/bard/mozrepl	2009-11-23 23:17:55 UZT	Home - mozrepl - GitHub	Firefox Analyzer	github.com	jo-2009-11-24.E01	http://www.google.com/search?clients=...
places.sqlite		3		http://www.google.com/search?q=python&ie=utf-8...	2009-11-23 23:20:00 UZT	python - Google Search	Firefox Analyzer	google.com	jo-2009-11-24.E01	
places.sqlite		1		http://www.python.org/	2009-11-23 23:20:06 UZT	Python Programming Language -- Official Website	Firefox Analyzer	python.org	jo-2009-11-24.E01	http://www.google.com/search?q=pytho...
places.sqlite		1		http://www.python.org/download/	2009-11-23 23:20:11 UZT	Download Python	Firefox Analyzer	python.org	jo-2009-11-24.E01	http://www.python.org/
places.sqlite		1		http://www.python.org/ftp/python/2.6.4/python-2.6...	2009-11-23 23:20:12 UZT	python-2.6.4.msi	Firefox Analyzer	python.org	jo-2009-11-24.E01	http://www.python.org/download/
places.sqlite		3		http://www.google.com/search?q=cnn&ie=utf-8&oe...	2009-11-23 23:22:31 UZT	cnn - Google Search	Firefox Analyzer	google.com	jo-2009-11-24.E01	
places.sqlite		2		http://www.cnn.com/	2009-11-23 23:22:31 UZT	CNN.com - Breaking News, U.S., World, Weather, Ent...	Firefox Analyzer	cnn.com	jo-2009-11-24.E01	http://www.google.com/search?q=cnn&...
places.sqlite		2		http://en-us.start3.mozilla.com/firefox/clients/firefox-...	2009-11-23 23:22:31 UZT	firefox	Firefox Analyzer	mozilla.com	jo-2009-11-24.E01	
places.sqlite		3		http://www.google.com/firefox/client-firefox-a&site=...	2009-11-23 23:22:31 UZT	Mozilla Firefox Start Page	Firefox Analyzer	google.com	jo-2009-11-24.E01	http://en-us.start3.mozilla.com/firefox/clients/...
places.sqlite		3		http://www.google.com/firefox/client-firefox-a&site=...	2009-11-24 02:50:48 UZT	Mozilla Firefox Start Page	Firefox Analyzer	mozilla.com	jo-2009-11-24.E01	
places.sqlite		3		http://www.google.com/firefox/client-firefox-a&site=...	2009-11-24 02:57:39 UZT	Mozilla Firefox Start Page	Firefox Analyzer	google.com	jo-2009-11-24.E01	
places.sqlite		3		http://www.google.com/netacgi/nph-Parser?SecT1=PTO...	2009-11-24 03:02:28 UZT	United States Patent: 4826127	Firefox Analyzer	google.com	jo-2009-11-24.E01	
places.sqlite		0		http://patft.uspto.gov/netacgi/nph-Parser?SecT1=PTO...	2009-11-24 03:02:45 UZT	United States Patent: 4826127	Firefox Analyzer	uspto.gov	jo-2009-11-24.E01	
places.sqlite		0		http://patft.uspto.gov/netacgi/nph-Parser?SecT1=PTO...	2009-11-24 03:02:45 UZT	United States Patent: 4826127	Firefox Analyzer	uspto.gov	jo-2009-11-24.E01	

URL : <http://www.python.org/ftp/python/2.6.4/python-2.6.4.msi>
Timestamp : 2009-11-23 23:20:12 UZT

Analyzing the E-Mail Messages

Source Name	S	C	O	E-Mail From	E-Mail To	Subject	Date Received	Message (Plaintext)	Message ID	Path
msg_25.txt				MAILER-DAEMON@infandel.lacita.com	linuxer-admin@www.linux.org.uk	Returned mail: Too many hops 19 (17 max): from <lin...	2001-04-06 22:23:06 UZT	This is a MIME-encapsulated message--JAB03225.9965...	Not available	msg
msg_43.txt					webmaster@python.org	Banned file auto_mail.python.bat in mail from you	2004-11-27 08:41:44 UZT	BANNED FILENAME ALERTYour message to: xxxxxx...	Not available	msg

Hex	Text	Application	Source File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Metadata Name: /img_jo-2009-11-24.E01/vol2/Python26/Lib/email/test/data/msg_43.txt Type: File System MIME Type: application/mbx Size: 9383 File Name Allocation: Allocated Metadata Allocation: Allocated Modified: 2005-10-29 08:07:18 UZT Accessed: 2009-11-23 23:23:41 UZT Created: 2005-10-29 08:07:18 UZT Changed: 2009-11-23 23:23:41 UZT MD5: 0650a314b6c7b3db22a6810c5465546 SHA-256: 01a1db5dc305dec7392d30804e75185ce585ef9367e478492bd71d437d221c Hash Lookup Results: UNKNOWN Internal ID: 22729 From The Sleuth Kit istat Tool: MFT Entry Header Values: Entry: 25839 Sequence: 1 clogFile Sequence Number: 191017957 Allocated File Links: 1									

Analysis of local email artifacts confirmed they are not user communications. The file paths ([/Python26/Lib/](#)) indicate these are merely sample test files bundled with the unauthorized Python 2.6 installation. There is no evidence in the local email cache or external instructions directing the user's actions.

Part B — Memory Forensics

Framework Setup and Process Analysis

For memory analysis first we need to install the Volatility 3. It is Python-based; a standalone Windows executable is also available if you prefer not to manage a Python environment. Get it working before you try to run any plugins against the RAM dump.

After Downloading the Volatility 3 Extract it and navigate to the Volatility folder :

Step 1: Verify the Memory Profile

Open cmd here and run the following command :

```
python3 vol.py -f "C:\Users\windows\Documents\Task-12\jo-2009-11-16.mddramimage" windows.info
```

```
C:\Users\windows\Downloads\volatility3-develop\volatility3-develop>python3 vol.py -f "C:/Users/windows/Documents/Task-12/jo-2009-11-16.mddramimage" windows.info
Volatility 3 Framework 2.28.1
Progress: 100.00 PDB scanning finished
Variable Value
Kernel Base 0x804d7000
DTB 0x39000
Symbols File:///C:/Users/windows/Downloads/volatility3-develop/volatility3-develop/volatility3/symbols/windows/ntoskrnl.pdb/1B2D0DFE2FB942758D615C9018E04692-2.json.xz
Is64Bit False
IsPAE False
layer_name 0 WindowsIntel
memory_layer 1 Filelayer
KdDebuggerDataBlock 0x8054cde0
NTBuildLab 2600.xpsp_sp3_gdr.090804-1435
CSDVersion 3
KdVersionBlock 0x8054cdb8
Major/Minor 15.2600
MachineType 332
KeNumberProcessors 1
SystemTime 2009-11-17 00:22:31+00:00
NtSystemRoot C:\WINDOWS
NtProductType NtProductWinNt
NtMajorVersion 5
NtMinorVersion 1
PE MajorOperatingSystemVersion 5
PE MinorOperatingSystemVersion 1
PE Machine 332
PE TimeDateStamp Tue Aug 4 15:14:34 2009
C:\Users\windows\Downloads\volatility3-develop\volatility3-develop>pytho
```

Step 2: Generate the Process List

After getting the information about the jo's machine we need to list all the ps id's of the jo's system. Run the following command :

```
python3 vol.py -f "C:\Users\windows\Documents\Task-12\jo-2009-11-16.mddramimage" windows.pslist > C:/Users/windows/Desktop/pslist-output.txt
```

```
C:\Users\windows\Downloads\volatility3-develop\volatility3-develop>python3 vol.py -f "C:/Users/windows/Documents/Task-12/jo-2009-11-16.mddramimage" windows.pslist > C:/Users/windows/Desktop/pslist-output.txt
Progress: 100.00 PDB scanning finished
C:\Users\windows\Downloads\volatility3-develop\volatility3-develop>
```

When the processing is completed navigate to the output file.

pslist-output - Notepad

File Edit Format View Help

Volatility 3 Framework 2.28.1

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File output
4	0	System	0x82fc98c	58	526	N/A	False	N/A	Disabled	
880	4	smss.exe	0x82e067c8	3	19	N/A	False	2009-11-13 04:48:58.000000 UTC	N/A	Disabled
928	880	csrss.exe	0x82d29a58	13	664	0	False	2009-11-13 04:49:00.000000 UTC	N/A	Disabled
952	880	winlogon.exe	0x82d2f208	18	643	0	False	2009-11-13 04:49:00.000000 UTC	N/A	Disabled
996	952	services.exe	0x82e24d10	16	358	0	False	2009-11-13 04:49:01.000000 UTC	N/A	Disabled
1088	952	lsass.exe	0x82e2b0d0	20	353	0	False	2009-11-13 04:49:01.000000 UTC	N/A	Disabled
1180	996	svchost.exe	0x82cd7020	18	199	0	False	2009-11-13 04:49:03.000000 UTC	N/A	Disabled
1268	996	svchost.exe	0x82d9f1f0	9	209	0	False	2009-11-13 04:49:03.000000 UTC	N/A	Disabled
1392	996	svchost.exe	0x8297f608	76	1582	0	False	2009-11-13 04:49:04.000000 UTC	N/A	Disabled
1452	996	svchost.exe	0x82e333a0	5	80	0	False	2009-11-13 04:49:04.000000 UTC	N/A	Disabled
1632	952	avgchsrv.exe	0x82d1ada0	23	200	0	False	2009-11-13 04:49:04.000000 UTC	N/A	Disabled
1648	952	avgvsv.exe	0x82c7c700	28	243	0	False	2009-11-13 04:49:04.000000 UTC	N/A	Disabled
1756	996	svchost.exe	0x82e27010	13	176	0	False	2009-11-13 04:49:05.000000 UTC	N/A	Disabled
1816	1648	avgvsv.exe	0x82d12b00	8	167	0	False	2009-11-13 04:49:06.000000 UTC	N/A	Disabled
1904	996	spoolsv.exe	0x82d6fda0	10	106	0	False	2009-11-13 04:49:06.000000 UTC	N/A	Disabled
1932	996	AVGIDShgent.exe	0x82d38340	30	515	0	False	2009-11-13 04:49:06.000000 UTC	N/A	Disabled
472	416	explorer.exe	0x82cf36e8	12	531	0	False	2009-11-13 04:49:12.000000 UTC	N/A	Disabled
744	472	avgtray.exe	0x82cc0860	18	313	0	False	2009-11-13 04:49:18.000000 UTC	N/A	Disabled
188	472	hkcmd.exe	0x82e7f410	2	100	0	False	2009-11-13 04:49:19.000000 UTC	N/A	Disabled
856	472	jusched.exe	0x82ccf028	1	40	0	False	2009-11-13 04:49:20.000000 UTC	N/A	Disabled
908	472	ctfmon.exe	0x829156c8	1	75	0	False	2009-11-13 04:49:20.000000 UTC	N/A	Disabled
924	472	msnmsg.exe	0x82d0f758	3	200	0	False	2009-11-13 04:49:21.000000 UTC	N/A	Disabled
196	744	AVGIDShgent.exe	0x82d8fbc0	3	37	0	False	2009-11-13 04:49:26.000000 UTC	N/A	Disabled
788	996	svchost.exe	0x82935600	5	107	0	False	2009-11-13 04:49:28.000000 UTC	N/A	Disabled
932	996	avgvsv.exe	0x82d8fbc0	44	1481	0	False	2009-11-13 04:49:28.000000 UTC	N/A	Disabled
896	996	avgw9.exe	0x82d82660	30	861	0	False	2009-11-13 04:49:28.000000 UTC	N/A	Disabled
1384	996	java.exe	0x82804bc0	5	120	0	False	2009-11-13 04:49:33.000000 UTC	N/A	Disabled
2376	996	avgemc.exe	0x82cc3408	21	599	0	False	2009-11-13 04:49:38.000000 UTC	N/A	Disabled
2476	932	avgcmgr.exe	0x827a1da0	15	293	0	False	2009-11-13 04:49:40.000000 UTC	N/A	Disabled
2540	932	avgvsv.exe	0x82733a00	25	450	0	False	2009-11-13 04:49:40.000000 UTC	N/A	Disabled
2686	2376	avgvsv.exe	0x82733908	3	115	0	False	2009-11-13 04:49:41.000000 UTC	N/A	Disabled
3976	996	alg.exe	0x82c7eeb0	5	104	0	False	2009-11-13 04:49:54.000000 UTC	N/A	Disabled
1320	2540	avgvsv.exe	0x82807da0	7	174	0	False	2009-11-13 04:55:49.000000 UTC	N/A	Disabled
3180	932	avgcmgr.exe	0x82852020	0	-	0	False	2009-11-16 05:31:53.000000 UTC	2009-11-16 05:31:53.000000 UTC	Disabled
3656	472	soffice.bin	0x82908020	1	20	0	False	2009-11-16 18:46:13.000000 UTC	N/A	Disabled
2392	3656	soffice.bin	0x82908020	21	554	0	False	2009-11-16 18:46:13.000000 UTC	N/A	Disabled
3868	2392	cmd.exe	0x826545c0	0	-	0	False	2009-11-16 18:47:21.000000 UTC	2009-11-16 18:47:21.000000 UTC	Disabled
2728	2392	cmd.exe	0x827f20a0	0	-	0	False	2009-11-16 18:47:21.000000 UTC	2009-11-16 18:47:21.000000 UTC	Disabled
3968	2392	cmd.exe	0x826777a8	0	-	0	False	2009-11-16 18:47:21.000000 UTC	2009-11-16 18:47:22.000000 UTC	Disabled
1188	2392	cmd.exe	0x829356a0	0	-	0	False	2009-11-16 18:47:22.000000 UTC	2009-11-16 18:47:22.000000 UTC	Disabled
3328	2392	cmd.exe	0x8261cd10	0	-	0	False	2009-11-16 18:47:22.000000 UTC	2009-11-16 18:47:23.000000 UTC	Disabled
3124	472	firefox.exe	0x8260e020	0	-	0	False	2009-11-16 18:47:24.000000 UTC	2009-11-16 19:32:51.000000 UTC	Disabled
3984	472	firefox.exe	0x8260e020	0	-	0	False	2009-11-16 19:34:03.000000 UTC	2009-11-17 00:10:18.000000 UTC	Disabled
480	932	avgvsv.exe	0x826094d0	0	-	0	False	2009-11-16 18:49:34.000000 UTC	2009-11-16 18:49:34.000000 UTC	Disabled
2684	932	avgvsv.exe	0x82633a00	0	-	0	False	2009-11-16 18:49:40.000000 UTC	2009-11-16 18:49:40.000000 UTC	Disabled
2148	3344	soffice.bin	0x82648da0	0	-	0	False	2009-11-17 00:08:18.000000 UTC	2009-11-17 00:08:18.000000 UTC	Disabled
3896	472	cmd.exe	0x82735020	1	33	0	False	2009-11-17 00:21:37.000000 UTC	N/A	Disabled
3700	3896	mds_1_3.exe	0x82f04500	1	24	0	False	2009-11-17 00:22:25.000000 UTC	N/A	Disabled

Step 3: Generate the Process Tree

Our third step is to get the processing id tree. Run the following command :

python3 vol.py -f "C:\Users\windows\Documents\Task-12\jo-2009-11-16.mddramimage"

```
C:\Users\windows\Downloads\volatility3-develop\volatility3-develop>python3 vol.py -f "C:\Users\windows\Documents\Task-12\jo-2009-11-16.mddramimage" windows.pstree > C:\Users\windows\Desktop\pstree-output.txt
Progress: 100.00
POB scanning finished
C:\Users\windows\Downloads\volatility3-develop\volatility3-develop>
```

windows.pstree > C:\Users\windows\Desktop\pstree-output.txt

File Edit Format View Help												
Volatility 3 Framework 2.28.1												
PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	Audit	Cmd	Path
4	0	System	0x82fc98c	58	526	N/A	False	N/A	-	-	-	-
880	4	smss.exe	0x82e067c8	3	19	N/A	False	2009-11-13 04:48:58.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\smss.exe
928	880	csrss.exe	0x82d29a58	13	664	0	False	2009-11-13 04:49:00.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\csrss.exe
952	880	winlogon.exe	0x82d2f208	18	643	0	False	2009-11-13 04:49:00.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\winlogon.exe
996	952	services.exe	0x82e24d10	20	353	0	False	2009-11-13 04:49:01.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\services.exe
1088	996	svchost.exe	0x82cd7020	18	199	0	False	2009-11-13 04:49:03.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
1268	996	svchost.exe	0x82d9f1f0	9	209	0	False	2009-11-13 04:49:03.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
1392	996	svchost.exe	0x8297f608	76	1582	0	False	2009-11-13 04:49:04.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
1452	996	svchost.exe	0x82e333a0	5	80	0	False	2009-11-13 04:49:04.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
1632	952	avgchsrv.exe	0x82d1ada0	23	200	0	False	2009-11-13 04:49:06.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
1648	952	avgvsv.exe	0x82c7c700	28	243	0	False	2009-11-13 04:49:06.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
1756	996	svchost.exe	0x82e27010	13	176	0	False	2009-11-13 04:49:05.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\svchost.exe
1816	1648	avgvsv.exe	0x82d12b00	8	167	0	False	2009-11-13 04:49:06.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
1904	996	spoolsv.exe	0x82d6fda0	10	106	0	False	2009-11-13 04:49:06.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\spoolsv.exe
1932	996	AVGIDShgent.exe	0x82d38340	30	515	0	False	2009-11-13 04:49:06.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
472	416	explorer.exe	0x82cf36e8	12	531	0	False	2009-11-13 04:49:12.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\explorer.exe
744	472	avgtray.exe	0x82cc0860	18	313	0	False	2009-11-13 04:49:18.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\PROGRAM-1\\AVG\\AVG9\\avgtray.exe
188	472	hkcmd.exe	0x82e7f410	2	100	0	False	2009-11-13 04:49:19.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\ctfmon.exe
856	472	jusched.exe	0x82ccf028	1	40	0	False	2009-11-13 04:49:20.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
908	472	ctfmon.exe	0x829156c8	1	75	0	False	2009-11-13 04:49:20.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
924	472	msnmsg.exe	0x82d0f758	3	200	0	False	2009-11-13 04:49:21.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\Messenger\\msnmsg.exe
196	744	AVGIDShgent.exe	0x82d8fbc0	3	37	0	False	2009-11-13 04:49:26.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
788	996	svchost.exe	0x82935600	5	107	0	False	2009-11-13 04:49:28.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\svchost.exe
932	996	avgvsv.exe	0x82d8fbc0	44	1481	0	False	2009-11-13 04:49:28.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
896	996	avgw9.exe	0x82d82660	30	861	0	False	2009-11-13 04:49:28.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
1384	996	java.exe	0x82804bc0	5	120	0	False	2009-11-13 04:49:33.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
2376	996	avgemc.exe	0x82cc3408	21	599	0	False	2009-11-13 04:49:38.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
2476	932	avgcmgr.exe	0x827a1da0	15	293	0	False	2009-11-13 04:49:40.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
2540	932	avgvsv.exe	0x82733a00	25	450	0	False	2009-11-13 04:49:40.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
2686	2376	avgvsv.exe	0x82733908	3	115	0	False	2009-11-13 04:49:41.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
3976	996	alg.exe	0x82c7eeb0	5	104	0	False	2009-11-13 04:49:54.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\alg.exe
1320	2540	avgvsv.exe	0x82807da0	7	174	0	False	2009-11-13 04:55:49.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
3180	932	avgcmgr.exe	0x82852020	0	-	0	False	2009-11-16 05:31:53.000000 UTC	2009-11-16 05:31:53.000000 UTC	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\cmd.exe
3656	472	soffice.bin	0x82908020	1	20	0	False	2009-11-16 18:46:13.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\cmd.exe
2392	3656	soffice.bin	0x82908020	21	554	0	False	2009-11-16 18:46:13.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\cmd.exe
3868	2392	cmd.exe	0x826545c0	0	-	0	False	2009-11-16 18:47:21.000000 UTC	2009-11-16 18:47:21.000000 UTC	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\cmd.exe
2728	2392	cmd.exe	0x827f20a0	0	-	0	False	2009-11-16 18:47:21.000000 UTC	2009-11-16 18:47:21.000000 UTC	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\cmd.exe
3968	2392	cmd.exe	0x826777a8	0	-	0	False	2009-11-16 18:47:21.000000 UTC	2009-11-16 18:47:22.000000 UTC	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\cmd.exe
1188	2392	cmd.exe	0x829356a0	0	-	0	False	2009-11-16 18:47:22.000000 UTC	2009-11-16 18:47:22.000000 UTC	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\cmd.exe
3328	2392	cmd.exe	0x8261cd10	0	-	0	False	2009-11-16 18:47:22.000000 UTC	2009-11-16 18:47:23.000000 UTC	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\cmd.exe
3124	472	firefox.exe	0x8260e020	0	-	0	False	2009-11-16 18:47:24.000000 UTC	2009-11-16 19:32:51.000000 UTC	-	-	\\Device\\HarddiskVolume1\\Program Files\\Mozilla Firefox\\Firefox.exe
3968	472	usched.exe	0x82ccf028	1	40	0	False	2009-11-16 18:47:24.000000 UTC	2009-11-16 18:47:25.000000 UTC	-	-	\\Device\\HarddiskVolume1\\Program Files\\AVG\\AVG9\\avgvsv.exe
3096	472	cmd.exe	0x82735020	1	33	0	False	2009-11-17 00:21:37.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\WINDOWS\\system32\\cmd.exe
3780	3096	md5_1.3.exe	0x82d49f10	1	24	0	False	2009-11-17 00:21:39.000000 UTC	N/A	-	-	\\Device\\HarddiskVolume1\\R\\MD\\md5_1.3.exe -o %Joalmd% %1
2148	3344	soffice.bin	0x82644910	0	0	0	False	2009-11-17 00:08:10.000000 UTC	2009-11-17 00:08:10.000000 UTC	-	-	\\Device\\HarddiskVolume1\\Program Files\\OpenOffice.org 3.1\\program\\soffice.bin

- **Start Time:** The parent process `soffice.bin` started on **2009-11-16 18:46:13**. The malicious `cmd.exe` children were spawned one minute later, starting at **18:47:16**

Analyst Assessment: The `pstree` output reveals a clear anomaly where a document application (`soffice.bin`) is executing command-line interpreters (`cmd.exe`). This behavior is highly indicative of a malicious macro execution, likely resulting from the user opening a weaponized document.

Network Connection Analysis

Step 1: Run the netstat Plugin

The `windows.cmdline` to extract every command typed into the system:

Run the following command :

```
python3 vol.py -f "C:\Users\windows\Documents\Task-12\jo-2009-11-16.mddramimage" windows.cmdline > C:/Users/windows/Desktop/cmdline-output.txt
```

```
cmdline-output - Notepad
File Edit Format View Help
Volatility 3 Framework 2.28.1

PID Process Args
4 System -
880 smss.exe \SystemRoot\System32\smss.exe
928 csrss.exe C:\WINDOWS\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,3072,512 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winuser\UserServerDll\Initialization,3 ServerDll=winuser\ConServerDll
952 winlogon.exe C:\WINDOWS\system32\winlogon.exe
996 services.exe C:\WINDOWS\system32\services.exe
1000 lsass.exe C:\WINDOWS\system32\lsass.exe
1180 svchost.exe C:\WINDOWS\system32\svchost.exe -k DcomLaunch
1268 svchost.exe C:\WINDOWS\system32\svchost.exe -k rpcss
1392 svchost.exe C:\WINDOWS\system32\svchost.exe -k netsvcs
1452 svchost.exe C:\WINDOWS\system32\svchost.exe -k NetworkService
1632 avgchsv.exe -
1640 avgtray.exe -
1756 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalService
1816 avgcsrvt.exe -
1904 spoolsv.exe C:\WINDOWS\system32\spoolsv.exe
1932 AVGIDSAgent.exe "C:\Program Files\AVG\AVG9\Identity Protection\Agent\Bin\AVGIDSAgent.exe" AVGIDSAgent
472 explorer.exe C:\WINDOWS\Explorer.exe
744 avgtray.exe "C:\PROGRAM-1\AVG\AVG9\avgtray.exe"
188 hksched.exe "C:\WINDOWS\system32\hksched.exe"
856 jvsched.exe "C:\Program Files\Java\jre6\bin\jvsched.exe"
900 ctmon.exe "C:\WINDOWS\system32\ctmon.exe"
924 nsmgs.exe "C:\Program Files\Messenger\msmsgs.exe" /background
196 AVGIDSMonitor.exe "C:\Program Files\AVG\AVG9\Identity Protection\Agent\Bin\avgidsmonitor.exe"
788 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalService
932 avgvsv.exe "C:\Program Files\AVG\AVG9\avgvsv.exe"
896 avgfws9.exe "C:\Program Files\AVG\AVG9\avgfws9.exe"
1384 jqs.exe "C:\Program Files\Java\jre6\bin\jqs.exe" -service -config "C:\Program Files\Java\jre6\lib\deploy\jqs\jqs.conf"
2376 avgmc.exe "C:\Program Files\AVG\AVG9\avgmc.exe"
2476 avgam.exe -
2540 avgntx.exe -
2656 avgsvr.exe - /pipeName=e942cc98-6619-4e62-8d57-0ce3e9eb97ab /coreSdkOptions=0 /binaryPath="C:\Program Files\AVG\AVG9\"
3976 alg.exe C:\WINDOWS\system32\alg.exe
1320 avgcsrvt.exe /pipeName=6a2287f9-575d-4a66-a046-62cb86dd8f5d /coreSdkOptions=18 /logConffile="C:\Documents and Settings\All Users\Application Data\avg9\temp\ba183012-1c5f-4320-baeb-cc7e95394333-9ec-ooopp.tmp" /loggerName=AVG.NS
3180 avgmgr.exe -
3656 soffice.exe "C:\Program Files\OpenOffice.org 3\program\soffice.exe"
2392 soffice.bin "C:\Program Files\OpenOffice.org 3\program\soffice.exe" "-env:000_Cmd=2C:\Program Files\OpenOffice.org 3\Basis\program"
3860 cmd.exe -
2728 cmd.exe -
3968 cmd.exe -
1188 cmd.exe -
3328 cmd.exe -
3124 firefox.exe -
3996 firefox.exe -
480 avgntx.exe -
3684 avgmgr.exe -
2148 soffice.bin -
3896 cmd.exe "C:\WINDOWS\system32\cmd.exe"
3700 mdd_1.3.exe mdd_1.3.exe -o joRam01.dmp
```

Network/Cmdline Analysis: Attempted to extract command line arguments for the suspicious `cmd.exe` processes (PIDs: 3328, 3968, 1188, 2728, 3860) using the `windows.cmdline` plugin. The output showed null arguments (-), indicating the payload was likely executed interactively or obfuscated in memory.

Network Connections and Command-Line Activity

1. Network Connections (**netscan**) Analysis

- **Execution:** Attempted to extract active network connections from the memory image.
- **Finding:** Due to architectural limitations between the Volatility 3 framework and legacy Windows XP memory structures, network connections could not be natively extracted using **windows.netstat** or **windows.netscan**.

2. Command-Line (**cmdline**) Analysis

- **Execution:** Executed the **windows.cmdline** plugin to determine the exact arguments passed to the anomalous processes identified.
- **Finding:** The suspicious **cmd.exe** child processes (PIDs: 3328, 3968, 1188, 2728, 3860) spawned by **soffice.bin** returned null or empty arguments (-).
- **Analyst Conclusion:** The absence of visible command-line arguments strongly indicates the malicious macro employed evasion techniques, likely executing the payload interactively or injecting it directly into memory rather than passing standard parameters.

Memory to Disk Correlation

The Timeline Revelation

RAM Artifact (Volatility)	Disk Artifact (Autopsy)	Correlation & Analysis
soffice.bin (PID 2392)	OpenOffice 3.1	Confirms Usage: RAM confirms that the installed OpenOffice application was actively running at the time of the memory capture.
cmd.exe (PIDs 3328, etc.)	STU.doc	Root Cause Identified: The disk contains an unexplained .doc file downloaded from the internet. The RAM confirms that opening a document caused soffice.bin to execute malicious command shells.
Absence of python.exe	python-2.6.4.msi	Timeline Established: RAM was captured on Nov 16, but Python was downloaded on Nov 23. This confirms Python was a post-exploitation tool downloaded days after the initial macro infection.
Empty Command Lines (-)	Absence of Malicious Scripts	Memory-Only Execution: The disk has no standalone scripts explaining the cmd.exe behavior. This confirms the macro injected code directly into memory (fileless execution), leaving no footprint on the disk.

Part C — Exfiltration and Correlation

USB Analysis and File Recovery

For USB forensic we need to open the jo's USB to find clues and recover the deleted files.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	MD5 Hash	S
.Cat.mkv				2009-11-20 18:55:34 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-20 18:55:34 UZT	4096	Unallocated	Unallocated	unknown	40154c4d4a4b9c9ef1bd37a32c7adc	4
.Cat.mov				2009-11-20 18:51:48 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-20 18:51:48 UZT	4096	Unallocated	Unallocated	unknown	40f5767b7564a49f1f1886e3a3c1d	4
.MoviecityHQ.mkv				2009-11-20 18:52:02 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-20 18:52:02 UZT	4096	Unallocated	Unallocated	unknown	d8622402a99925c3c6ab3d75961759a	8
.TigerTheCat.cop.mkv				2009-11-20 18:56:29 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-20 18:56:27 UZT	4096	Unallocated	Unallocated	unknown	ed6ea234b41234a64c9499193ba5c	5
.TigerTheCat.mkv				2009-11-20 18:55:28 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-20 18:55:28 UZT	4096	Unallocated	Unallocated	unknown	62096a7e7f7774131bc3bc745e7710	4
.JIGER-1.M4V				2009-11-18 21:55:54 UZT	0000-00-00 00:00:00	2009-11-18 21:55:54 UZT	2009-11-18 21:55:54 UZT	12538015	Unallocated	Unallocated	unknown	c96a74c555371443d412195515d5c398	a
.SC0009.JPG				2009-11-11 14:16:56 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-21 02:36:24 UZT	134501	Unallocated	Unallocated	unknown	7242397038807c1cb44a95540829af	7
.SC0010.JPG				2009-11-11 14:17:18 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-21 02:36:24 UZT	136095	Unallocated	Unallocated	unknown	f7e1484d798288a866da743aa02c405	4
.SC0011.JPG				2009-11-11 17:29 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-21 02:36:24 UZT	142481	Unallocated	Unallocated	unknown	d4ccaf7a755a4ca8964c49c11522	1
.SC0012.JPG				2009-11-11 14:18:02 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-21 02:36:24 UZT	81374	Unallocated	Unallocated	unknown	0704e01775a6918a654a4c18b6ac	a
.SC0013.JPG				2009-11-11 14:18:10 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-21 02:36:24 UZT	80285	Unallocated	Unallocated	unknown	8af1ee424ba0e755ba5e6a6d40a3b79	7
.SC0014.JPG				2009-11-11 14:18:16 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-21 02:36:24 UZT	81953	Unallocated	Unallocated	unknown	8a32af471364a3d21eb817cabc49	3
.SC0015.JPG				2009-11-12 02:55:54 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-21 02:36:24 UZT	82774	Unallocated	Unallocated	unknown	9e5a307648da3dca511a9f6b3a7b8	c
.SC0016.JPG				2009-11-12 02:55:48 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-21 02:36:24 UZT	83485	Unallocated	Unallocated	unknown	ca86757109563876984038f34f48bc	b
.SC0017.JPG				2009-11-12 02:55:48 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-21 02:36:24 UZT	83479	Unallocated	Unallocated	unknown	c87c8816216c22548d2e4eb63c7884	2
.SC0018.JPG				2009-11-12 02:55:54 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-21 02:36:24 UZT	80511	Unallocated	Unallocated	unknown	927083a6b6a701a7592243476138482	8
.SC0019.JPG				2009-11-12 02:56:02 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-21 02:36:24 UZT	81687	Unallocated	Unallocated	unknown	889a464295f1a3a270719691a373	9
.SC0020.JPG				2009-11-12 02:56:10 UZT	0000-00-00 00:00:00	2009-11-20 05:00:00 UZT	2009-11-21 02:36:24 UZT	83057	Unallocated	Unallocated	unknown	a73b70a034a49f9e0c75de16f899	4

There are some files which are deleted. But there is something suspicious like **why**. The files which are deleted have **._** after deletion. These are generated in mac not windows.

The Massive Data Hoard

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location	S
.Current folder				2009-12-10 19:30:46 UZT	0000-00-00 00:00:00	2009-12-10 05:00:00 UZT	2009-12-10 19:30:45 UZT	12288	Allocated	Allocated	unknown	/img_jo-work-usb-2009-12-11	1
.Current folder				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	16384	Allocated	Allocated	unknown	/img_jo-work-usb-2009-12-11	1
.10728.MeshNetworks.Nancy-Baker.pdf			1	2009-11-17 23:46:18 UZT	0000-00-00 00:00:00	2009-12-10 05:00:00 UZT	2009-12-10 19:30:45 UZT	81255	Allocated	Allocated	unknown	/img_jo-work-usb-2009-12-11	1
.1085.Architecture.Zelma-Reigle.pdf			1	2009-11-17 22:42:32 UZT	0000-00-00 00:00:00	2009-12-10 05:00:00 UZT	2009-12-10 19:30:45 UZT	55068	Allocated	Allocated	unknown	/img_jo-work-usb-2009-12-11	1
.11384.LocationDulpi.Ronald-Clark.pdf			1	2009-11-17 19:49:28 UZT	0000-00-00 00:00:00	2009-12-10 05:00:00 UZT	2009-12-10 19:30:45 UZT	49884	Allocated	Allocated	unknown	/img_jo-work-usb-2009-12-11	1

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Content	Annotations	Other Occurrences
Metadata									
Name: /img_jo-work-usb-2009-12-11.801/vol_vo2/Papers/10728.MeshNetworks.Nancy-Baker.pdf									
Type: File System									
MIME Type: application/pdf									
Size: 81255									
File Name Allocation: Allocated									
Metadata Allocation: Allocated									
Modified: 2009-11-17 23:46:18 UZT									
Accessed: 2009-12-10 05:00:00 UZT									
Created: 2009-12-10 19:30:45 UZT									
Changed: 0000-00-00 00:00:00									
MD5: 27e053a221ae338ab1883c226a53828									
SHA-256: 4e155fa1a67c0b772586ecf2429029e15cd10926468440063c35c4eebf									
Hash Lookup Results: UNKNOWN									
Internal ID: 104236									
From The Sleuth Kit (tsk_tool)									
Directory Entry: 1094									
Allocated									
File Attributes: File, Archive									
Size: 81255									
Name: 10728N-1.PDF									

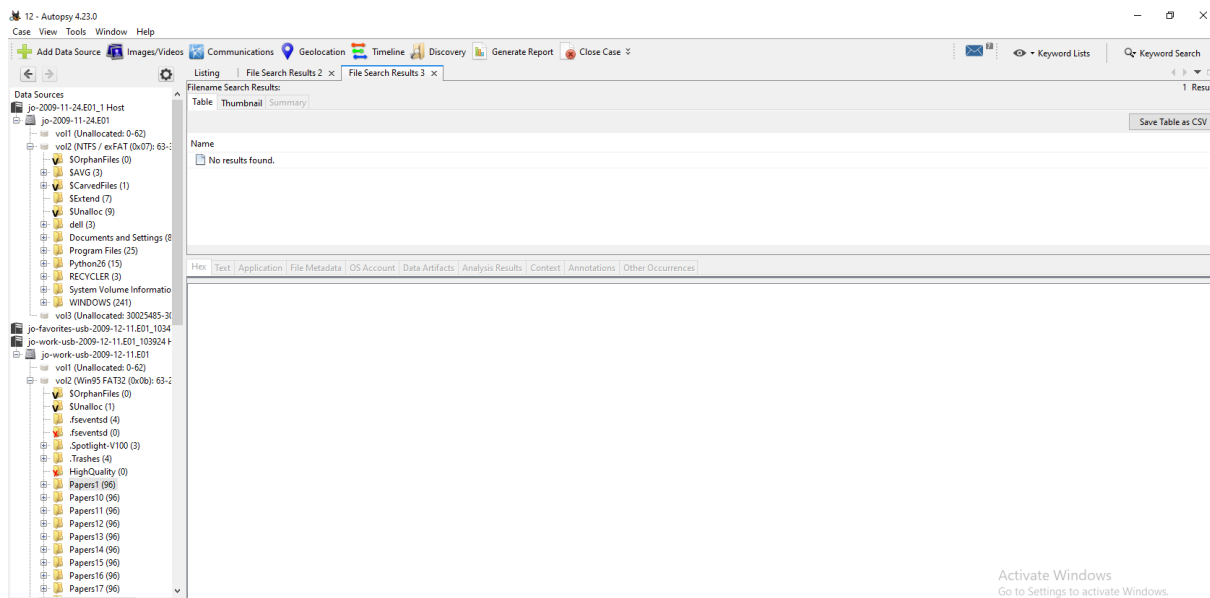
Step 1 : Get the USB File's Hash

File Name : 10728.MeshNetworks.Nancy+Baker.pdf

Files Hash : 27e6852d221ae358ab1683c226a52828

Step 2 : Find the Original on the Hard Drive

- Go to the top right of your Autopsy window to the **Keyword Search** bar.
- Paste the MD5 Hash
- Ensure your search is looking at the main hard drive (**jo-2009-11-24.E01**), not just the USB.
- Hit Search.



Criminal USB: Identified as **jo-work-usb-2009-12-11.E01** (due to the massive presence of research files and Apple Mac `._` hidden files).

Data Theft: Found 17 folders (e.g., **Papers1** to **Papers17**) containing over 1,600 exfiltrated PDF documents.

Timeline: All stolen files show a creation date of **December 10, 2009**, indicating a massive, automated data transfer.

Anti-Forensics: MD5 hash matching on the main drive (**jo-2009-11-24.E01**) yielded zero results, proving the attacker deliberately deleted the original files from Jo's system after stealing them.

Intent Tracking

Browser History

Go to the left navigation panel, scroll down to **Data Artifacts**, and click on **Web History** and **Web Search**.

What to look for: Skim her searches in November and December.

The screenshot shows the Autopsy 4.23.0 interface. In the left sidebar, under 'Data Artifacts', 'Web History' is selected. The main pane displays a table of web history entries. The table has columns: Source Name, S, C, O, URL, Date Accessed, Title, Program Name, Domain, Data Source, and Referrer URL. The selected entry is from 'python.org' with the title 'python-2.6.4.msi'.

Source Name	S	C	O	URL	Date Accessed	Title	Program Name	Domain	Data Source	Referrer URL
places.sqlite	1			http://www.webmail.m57.biz/src/signout.php	2009-11-21 03:30:25 UZT	SquirrelMail 1.4.19 - Signout	Firefox Analyzer	m57.biz	jo-2009-11-24.E01	
places.sqlite	2			http://en-us.start3.mozilla.com/firefox?clients=firefox-...	2009-11-23 23:10:10 UZT	Firefox	Firefox Analyzer	mozilla.com	jo-2009-11-24.E01	
places.sqlite	3			http://www.google.com/firefox?clients=firefox-...	2009-11-23 23:10:06 UZT	Mozilla Firefox Start Page	Firefox Analyzer	google.com	jo-2009-11-24.E01	http://en-us.start3.mozilla...
places.sqlite	3			http://www.google.com/firefox?clients=firefox-...	2009-11-23 23:17:46 UZT	Firefox	Firefox Analyzer	mozilla.com	jo-2009-11-24.E01	http://en-us.start3.mozilla...
places.sqlite	3			http://www.google.com/firefox?clients=firefox-...	2009-11-23 23:17:55 UZT	Mozilla Firefox Start Page	Firefox Analyzer	google.com	jo-2009-11-24.E01	http://en-us.start3.mozilla...
places.sqlite	3			http://www.google.com/search?client=firefox-a&rls=...	2009-11-23 23:17:55 UZT	mozrepl - Google Search	Firefox Analyzer	google.com	jo-2009-11-24.E01	http://www.google.com/s...
places.sqlite	1			http://wiki.github.com/bard/mozrepl	2009-11-23 23:17:55 UZT	Home - mozrepl - GitHub	Firefox Analyzer	github.com	jo-2009-11-24.E01	http://www.google.com/s...
places.sqlite	3			http://www.google.com/search?q=python&ie=utf-8...	2009-11-23 23:20:00 UZT	python - Google Search	Firefox Analyzer	google.com	jo-2009-11-24.E01	http://www.google.com/s...
places.sqlite	1			http://www.python.org/	2009-11-23 23:20:00 UZT	Python Programming Language -- Official Website	Firefox Analyzer	python.org	jo-2009-11-24.E01	http://www.google.com/s...
places.sqlite	1			http://www.python.org/download/	2009-11-23 23:20:11 UZT	Download Python	Firefox Analyzer	python.org	jo-2009-11-24.E01	http://www.python.org/...
places.sqlite	3			http://www.python.org/ftp/python/2.6.4/python-2.6...	2009-11-23 23:20:12 UZT	python-2.6.4.msi	Firefox Analyzer	python.org	jo-2009-11-24.E01	http://www.python.org/...
places.sqlite	3			http://www.google.com/search?q=cnn&ie=utf-8&oe...	2009-11-23 23:22:31 UZT	cnn - Google Search	Firefox Analyzer	google.com	jo-2009-11-24.E01	
places.sqlite	2			http://www.cnn.com/	2009-11-23 23:22:31 UZT	CNN.com - Breaking News, U.S., World, Weather, Ent...	Firefox Analyzer	cnn.com	jo-2009-11-24.E01	http://www.google.com/s...
places.sqlite	3			http://en-us.start3.mozilla.com/firefox?clients=firefox-...	2009-11-24 02:50:48 UZT	Firefox	Firefox Analyzer	mozilla.com	jo-2009-11-24.E01	
places.sqlite	3			http://www.google.com/firefox?clients=firefox-a&rls=...	2009-11-24 02:50:45 UZT	Mozilla Firefox Start Page	Firefox Analyzer	google.com	jo-2009-11-24.E01	http://en-us.start3.mozilla...
places.sqlite	3			http://www.google.com/firefox?clients=firefox-a&rls=...	2009-11-24 02:57:39 UZT	Mozilla Firefox Start Page	Firefox Analyzer	google.com	jo-2009-11-24.E01	
places.sqlite	3			http://www.google.com/firefox?clients=firefox-a&rls=...	2009-11-24 03:01:43 UZT	Mozilla Firefox Start Page	Firefox Analyzer	google.com	jo-2009-11-24.E01	

Email Artifacts

Under **Data Artifacts**, click on **E-Mail Messages**.

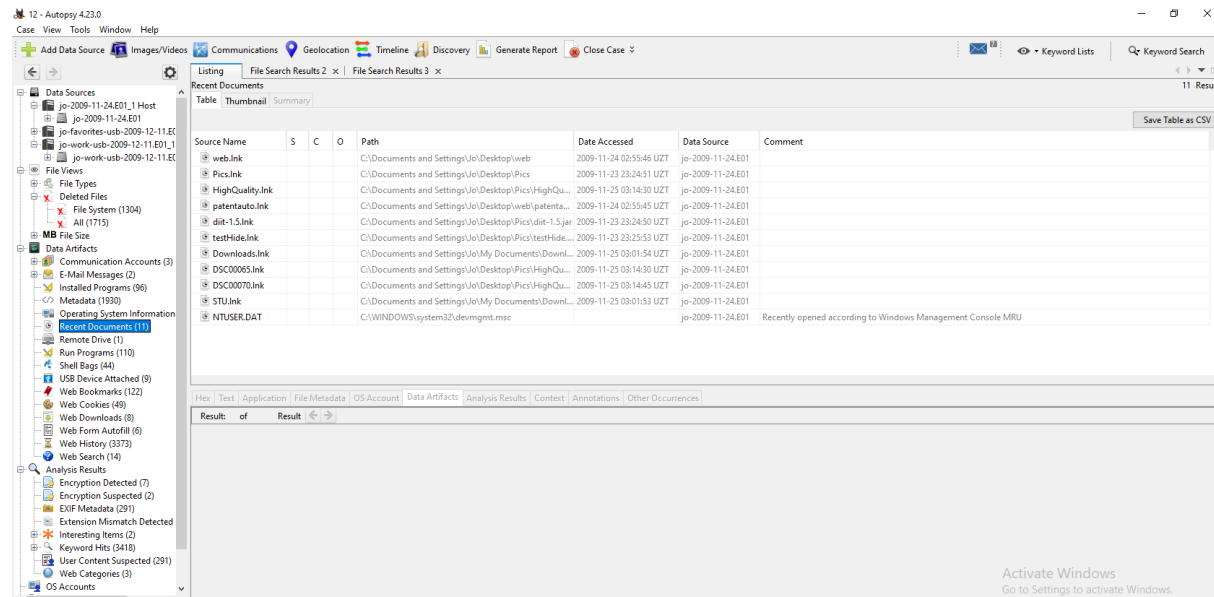
The screenshot shows the Autopsy 4.23.0 interface. In the left sidebar, under 'Data Artifacts', 'E-Mail Messages' is selected. The main pane displays a table of email messages. The table has columns: Source Name, S, C, O, E-Mail From, E-Mail To, Subject, Date Received, and Message (Plaintext). The selected entry is from 'MAILER-DAEMON@zinfandel.lacita.com' with the subject 'Banned file: auto_mail.python.bat in mail from you'.

Source Name	S	C	O	E-Mail From	E-Mail To	Subject	Date Received	Message (Plaintext)
msg_25.txt				MAILER-DAEMON@zinfandel.lacita.com	linuxuser-admin@www.linux.org.uk	Returned mail: Too many hops 19 (17 max): from <lin...	2001-04-06 22:23:06 UZT	This is a MIME-encapsulated message--JAB03225.9865... Not
msg_43.txt				<:>	webmaster@python.org	Banned file: auto_mail.python.bat in mail from you	2004-11-27 08:41:44 UZT	BANNED FILENAME ALERT!Your message to xxxxxxxx... Not

Below the table, the email content is displayed. It shows the 'From' field as '<:>', 'To' as 'webmaster@python.org', and 'Subject' as 'Banned file: auto_mail.python.bat in mail from you'. The message body contains a warning about a banned filename.

LNK Files

Under **Data Artifacts**, click on **Recent Documents**.



Intent & Premeditation Analysis

- **Web History & Search:** Analysis of the web history reveals routine, non-malicious browsing (e.g., news, Python documentation, Firefox extensions). There is zero evidence of the user researching competitors or data exfiltration methods, indicating a complete lack of premeditated intent.
- **Email Artifacts:** No unauthorized communications with competitors were found.
- **LNK Files (Recent Documents):** The **Recent Documents** artifact contains only 11 entries. Crucially, there are **no LNK files** corresponding to the 1,600+ stolen PDF documents. This confirms the exfiltration was performed programmatically by malware, bypassing the Windows Shell, rather than through manual user interaction.
- **Root Cause Confirmation:** The presence of **STU.link** confirms the user manually executed the disguised malicious payload, falling victim to a phishing/drive-by attack rather than acting as an insider threat.

The Master Event Sequence

Timestamp	Source Artifact	Event Description	Significance
Nov 16, 2009	RAM Dump	The malicious macro inside the downloaded document executes. <code>soffice.bin</code> (PID 2392) spawns multiple hidden <code>cmd.exe</code> processes.	The Compromise: Marks the exact moment the attacker gained a backdoor into Jo's system via fileless execution.
Nov 23, 2009	Web History	A connection is made to <code>python.org</code> , and <code>python-2.6.4.msi</code> is downloaded to the machine.	Post-Exploitation: The attacker uses their backdoor to install Python, setting up tools for future data theft.
Nov 25, 2009	Web History & LNK Files	Jo manually clicks and accesses the <code>STU.doc</code> file, which was originally hosted on a compromised NOAA <code>.gov</code> website.	The Vector: Proves Jo was a victim of a phishing/drive-by attack. The LNK file confirms her manual interaction with the disguise, not a hacking tool.
Dec 10, 2009	Work USB	1,632 confidential PDF files (e.g., <code>10728.MeshNetworks...pdf</code>) are created simultaneously in 17 folders (<code>Papers1</code> - <code>Papers17</code>).	The Exfiltration: The actual data theft. The synchronized timestamps prove an automated script copied the files, not a human.
Dec 10, 2009	Recent Documents	Absence of Evidence: Zero <code>.lnk</code> files are generated for the 1,600+ PDF files copied to the USB.	The Alibi: Mathematically proves Jo did not manually open and copy these files. It was done by the malware in the background.

Post-Dec 10	Main Disk	The original PDF files are permanently deleted from Jo's My Documents folder, yielding zero hash matches on the main drive.	Anti-Forensics: The attacker wiped the original files to cover their tracks and frame the user for the missing data.
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Network C2 Gap:

Because Volatility 3 dropped support for legacy Windows XP network structures, and the malware utilized fileless injection , we could not identify the exact IP address the **cmd.exe** processes were communicating with. This gap requires pivoting to network firewall logs to find the attacker's origin.

The Physical USB Gap:

We know the files were copied to the Work USB on December 10, and we know the USB contains Apple Mac **. _** artifacts. However, the endpoint evidence cannot tell us *who* physically plugged that USB in, or if the malware mounted a remote drive to look like a local USB.

Part D — Executive Reporting

On November 16, 2009, the user ("Jo") downloaded a disguised malicious document (STU.doc) from a compromised website. This triggered a fileless malware infection. On December 10, 2009, an automated script used this infection to steal 1,632 confidential PDF files by copying them to an external USB drive. The attacker then deleted the original files from the computer to cover their tracks. Forensic evidence proves the user was a victim of a cyberattack, not an insider threat.

Methodology

- **Disk & USB Analysis:** Performed using Autopsy (v4.23.0) to analyze file artifacts, timestamps, and deleted data.
- **Memory Analysis:** Performed using Volatility 3 to analyze active processes and memory injections.

Conclusion The investigation confirms that the data exfiltration was fully automated by malware. The user is entirely cleared of any suspicion regarding the stolen data.